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Predicting coral bleaching severity using global survey & environmental data

IMPERIAL COLLEGE BUSINESS SCHOOL

PROFESSIONAL CERTIFICATE IN DATA ANALYTICS

Executive summary

This analysis investigates global coral bleaching patterns using integrated observational and environmental data, with the aim of identifying key drivers and improving understanding of bleaching risk. Results show that bleaching severity varies significantly across regions, indicating that local environmental conditions and reef resilience play an important role in shaping outcomes.

Thermal stress is identified as the primary driver of bleaching. In particular, cumulative heat exposure (measured as Degree Heating Weeks) and the frequency of thermal stress events are strongly associated with increased bleaching severity.

While linear models explain a modest proportion of variation (~12%), improved performance of tree-based models highlights the importance of non-linear interactions and threshold effects in driving bleaching dynamics. These findings emphasise the complexity of coral reef responses and the need for modelling approaches that capture these dynamics.

Overall, the results highlight the importance of incorporating both local context and advanced modelling techniques in predicting bleaching risk and informing conservation strategies.

The problem

Coral-reef bleaching is increasing in frequency, severity, and range as a result of climate change, posing a significant threat to reef ecosystems worldwide. Given the ecological, economic, and cultural importance of coral reefs, continued efforts to improve understanding of bleaching processes are critical. Analysing global bleaching survey data alongside environmental variables is necessary to better identify the conditions most strongly associated with bleaching under current and emerging climatic regimes, and to support more effective prediction, conservation, and management efforts.

Why is it important to solve?

Coral bleaching threatens the structure and function of reef ecosystems globally. Improving the ability to predict bleaching can strengthen conservation responses by enabling managers and organisations to:

- Identify sites at highest risk of bleaching
- Understand how warming trends influence reef vulnerability
- Prioritise monitoring efforts during periods of increasing heat stress
- Develop targeted mitigation strategies and restoration plans

Data description

The dataset used in this analysis is global coral bleaching and environmental data sourced from the [Biological and Chemical Oceanography Data Management Office \(BCO-DMO\)](#). It contains observations from coral reef survey sites worldwide collected between 1980 and 2020, providing a long-term perspective on bleaching patterns and associated environmental conditions.

The dataset comprises 41,361 records and 62 variables. Of these, 8 variables are numerical and 54 are categorical, capturing both quantitative environmental metrics and descriptive site characteristics. Key variables include the presence and extent of coral bleaching, site exposure, distance to land, mean turbidity, cyclone frequency, and multiple sea surface temperature metrics measured at the time of survey.

The target variable for this analysis is Percent_Bleaching, which quantifies the proportion of coral affected at each site and enables comparative analysis across locations and time periods. The breadth and temporal coverage of this dataset makes it well suited for investigating global bleaching trends and exploring relationships between thermal stress, environmental conditions, and bleaching severity.

Methodology

Data cleaning and preparation focused on ensuring data integrity through inspection, validation of data types, and identification and treatment of missing or inconsistent values.

Exploratory data analysis (EDA) used summary statistics and visual analysis to assess data quality, examine variable distributions, and explore key relationships between environmental factors and bleaching severity.

Data pre-processing included selecting relevant features, applying appropriate missing-data handling and scaling, and partitioning the dataset to support robust model evaluation.

Regression analysis compared a Dummy Regressor baseline with OLS, Random Forest, and XGBoost models to assess linear and non-linear relationships between environmental variables and percent bleaching.

Key challenges included incomplete environmental observations, differing variable scales, and an uneven distribution of bleaching severity, all addressed through careful preprocessing and model selection.

Exploratory Data Analysis

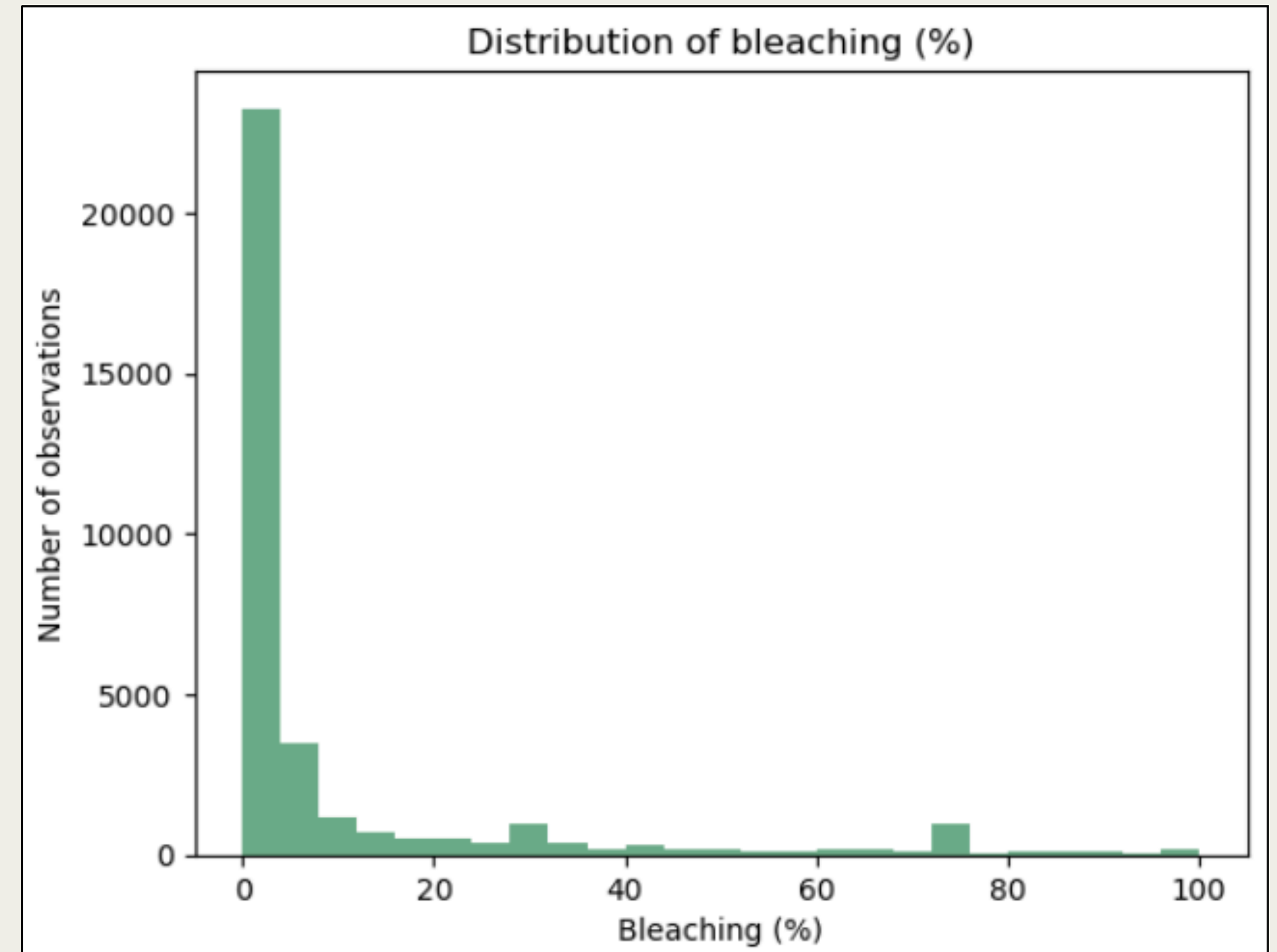
EDA was undertaken to understand the **structure, quality, and key characteristics of the coral bleaching dataset** prior to modelling. Initial analysis focused on data completeness and distributions, using summary statistics and visual exploration to identify missing values, skewed variables, and potential outliers that could influence results.

Univariate analysis was used to **examine the distribution of Percent_Bleaching across sites, years, and ocean basins**, revealing a highly skewed distribution with many zero-bleaching observations and episodic extreme events. Bivariate analysis **explored relationships between bleaching severity and key environmental variables**, particularly sea surface temperature metrics, as well as spatial and temporal patterns.

Spatial and temporal aggregation was also used to compare bleaching patterns across ocean regions and over time. This helped identify regional differences, trends in bleaching intensity, and heterogeneity in environmental exposure, directly informing feature selection and model choice in subsequent regression analysis.

Bleaching percent

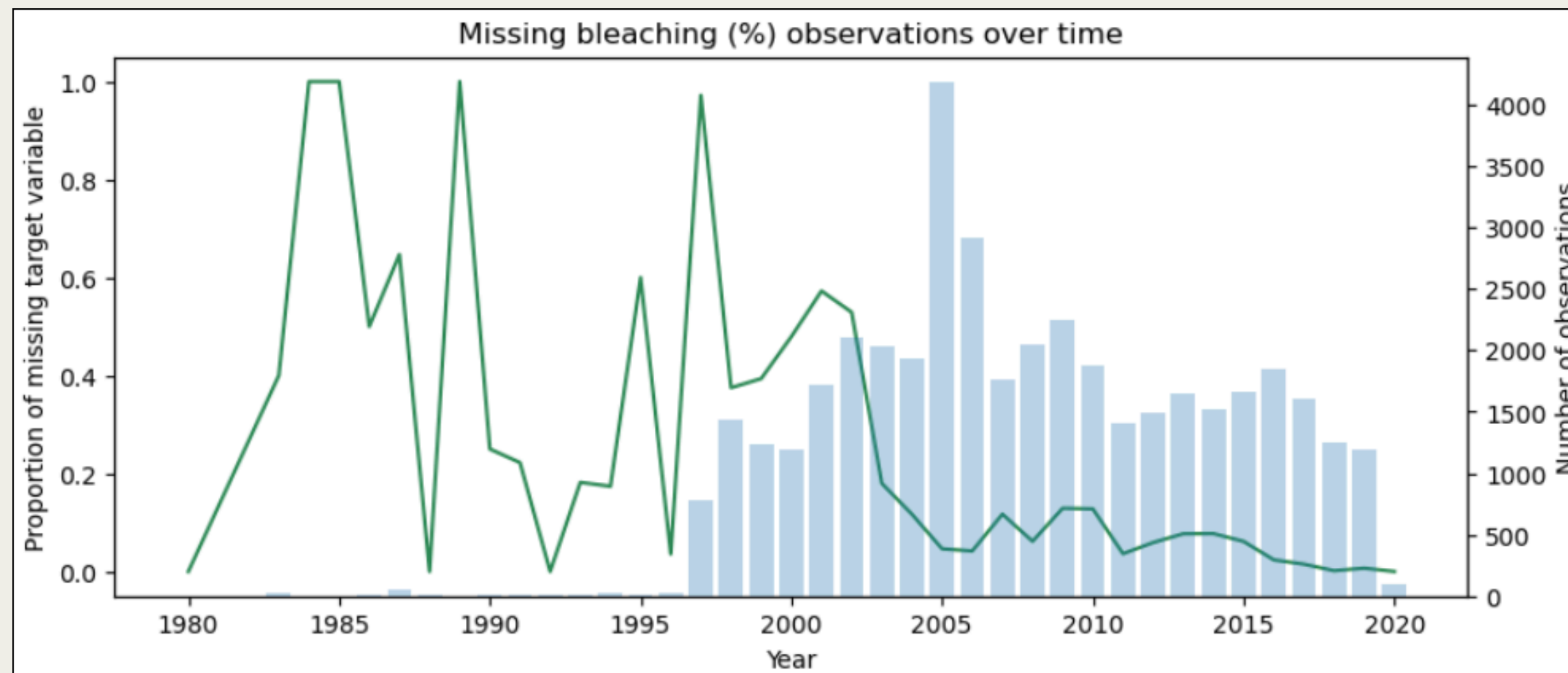
- A histogram of the bleaching (%) shows a clear a right skew
- Over 75% of the bleaching observations were between 0 - 8%
- The distribution indicates that most reefs experienced very low levels of bleaching
- A small number of observations experienced very high bleaching
- The mean of Percent_Bleaching will be influenced by these rare events, and the median will provide a more representative measure of typical conditions



Bleaching percent

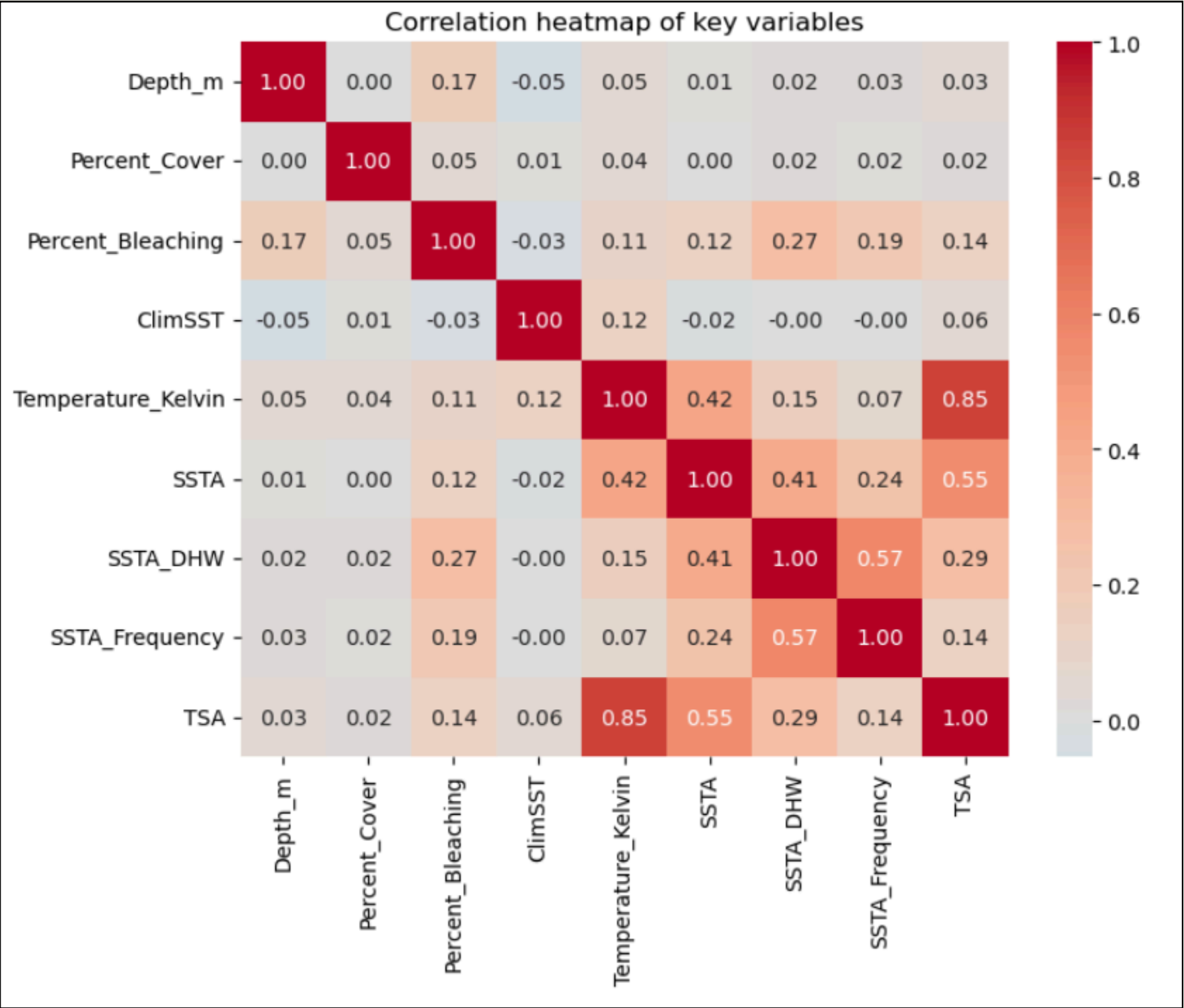
- **Bleaching (%) data was missing from 16% of observations**
- Missingness is higher between 1980 and early 2000s
- Years with 100% missing have a single or very few rows
- Missingness is random across regions

Rows missing the target variable cannot be used for model training, as the model cannot learn without a defined outcome. Imputing these values would introduce artificial bias, so observations with missing target variable were removed prior to machine-learning analysis.



Correlation heatmap

- The heatmap indicates that **Percent_Bleaching has generally weak linear relationships with most variables**, suggesting that bleaching cannot be explained by any single factor alone.
- The **strongest associations are with short-term temperature stress metrics**, particularly (SST_DWH) Degree Heating Weeks and the frequency of temperature anomalies, showing that bleaching tends to increase when elevated temperatures are sustained over time.
- In contrast, **long-term average sea surface temperature, depth, and percent coral cover show little to no correlation with bleaching**, indicating that baseline conditions and reef structural characteristics are less important than unusual heat stress events.
- Several temperature-related variables are strongly correlated with each other, reflecting that they capture similar aspects of thermal stress and **should be used carefully together in modelling**.
- Overall, the heatmap suggests that **coral bleaching in this dataset is primarily driven by rare but intense and prolonged heat events rather than by typical environmental conditions**.



Regional variation in bleaching

Analysing regional variation is **essential for understanding how coral bleaching responses differ across ocean basins**, rather than assuming a uniform global response to environmental stress. Environmental conditions, heat exposure, and local stressors vary substantially between regions, which can influence both bleaching severity and frequency.

To assess these differences, the dataset was grouped by ocean basin and regional bleaching severity was summarised using descriptive statistics, including mean, median, range, and observation counts. This enabled **comparison of typical bleaching levels and the occurrence of extreme events across regions**, while also highlighting differences in data coverage.

Regional patterns were then examined in relation to key environmental variables, particularly sea surface temperature metrics. Comparing these relationships across basins helped reveal heterogeneity in bleaching responses and informed more regionally nuanced interpretation of results.

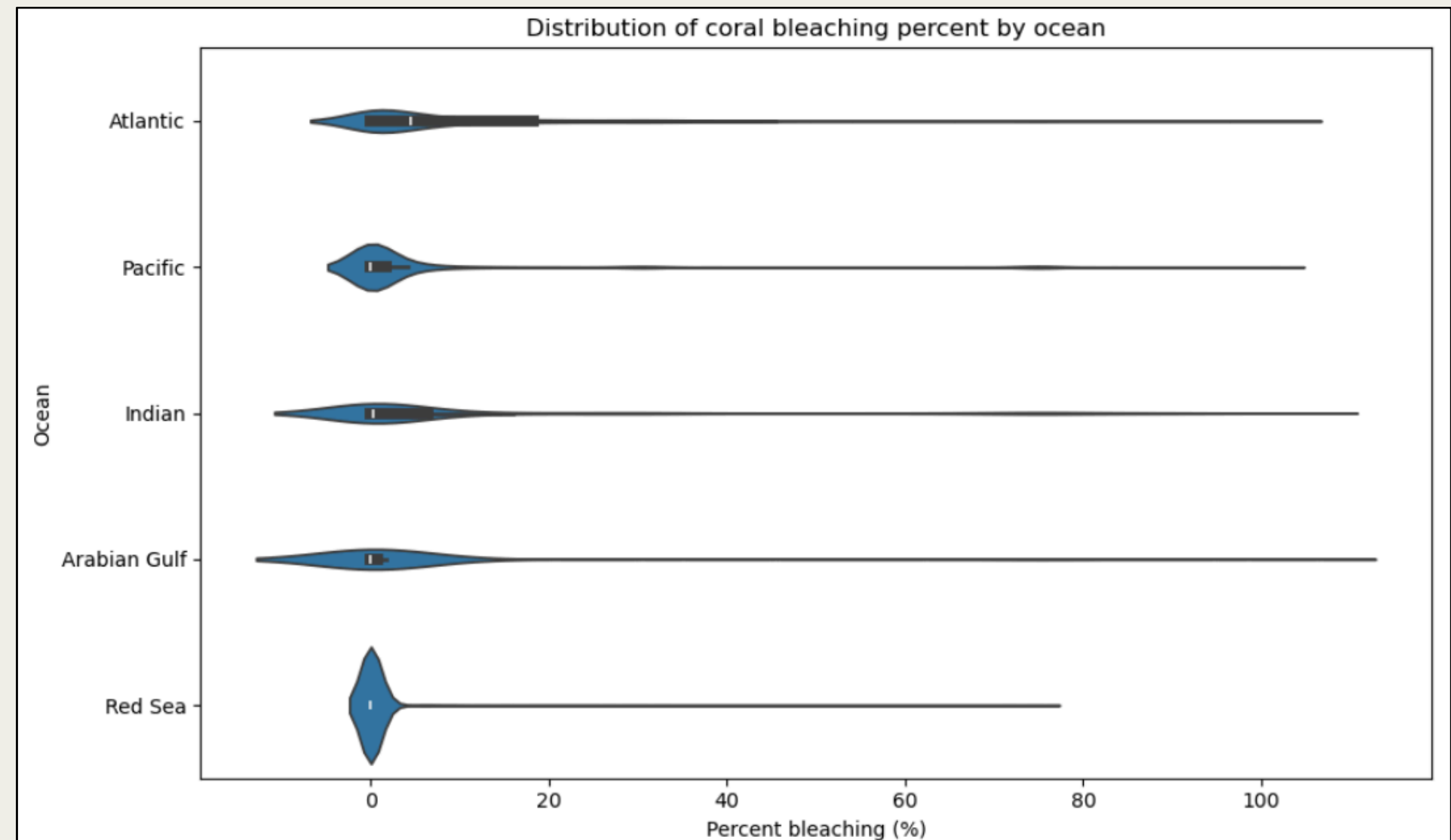
Regional variation in bleaching

	Ocean_Name	min_bleaching	max_bleaching	mean_bleaching	median_bleaching	observations
0	Arabian Gulf	0.0	100.0	6.68	0.00	369
1	Atlantic	0.0	100.0	14.40	4.46	13319
2	Indian	0.0	100.0	12.91	0.25	2327
3	Pacific	0.0	100.0	6.13	0.00	17446
4	Red Sea	0.0	75.0	0.66	0.00	1054

- **Bleaching severity varies substantially by ocean basin.** The Atlantic and Indian Oceans show the highest mean bleaching levels (mean ~ 14% and 13%), while the Pacific and Arabian Gulf have lower mean bleaching despite experiencing extreme events.
- **Extreme bleaching events occur across all regions**, with maximum bleaching reaching 100% in 4/5 oceans, however median bleaching is zero in 3/5 basins. This indicates bleaching is often episodic rather than persistent.
- **The Pacific dominates the dataset in coverage**, accounting for the largest number of observations (~17,400), suggesting that global bleaching patterns may be strongly influenced by Pacific reef dynamics and monitoring intensity.

Regional variation in bleaching

- Coral bleaching percentages show strong **right-skewed distributions across all oceans**, with most observations **clustered at low bleaching levels and a small number of extreme events**.
- The Pacific, Indian, Atlantic, and Arabian Gulf all exhibit long upper tails, indicating occasional severe bleaching events despite generally low median values.
- The Arabian Gulf displays **particularly high variability**, consistent with frequent exposure to extreme thermal conditions.
- The Red Sea shows a narrower distribution with generally lower bleaching percentages, though this may partly reflect lower observation density rather than reduced bleaching risk.



Data pre-processing

Feature–target definition: The target variable (Percent_Bleaching) was separated from the predictors to define the feature matrix (X) and response vector (y) prior to preprocessing and modelling.

Feature filtering and selection: Quality-based feature filtering was applied to remove redundant or non-informative variables, including identifier fields (e.g. site name) and features with substantial missingness and limited analytical value (e.g. Percent_Cover). These decisions were based on data quality, domain understanding, and exploratory analysis, rather than model performance.

Feature engineering: All predictors were checked for numerical suitability. The Exposure variable was ordinally encoded using a predefined, domain-driven mapping to reflect increasing environmental exposure. As this transformation was rule-based, it was applied consistently without risk of data leakage.

Train–test split: The dataset was split into training and test sets to enable unbiased evaluation of model performance and reduce overfitting risk.

Missing data handling: Missing values in remaining features were handled based on extent and relevance. Variables with low missingness (e.g. Depth, ~4%) were median-imputed using statistics calculated from the training data only and applied to both training and test sets to prevent data leakage.

Dummy Regressor

- RMSE: 19.91
- MAE: 13.15
- R^2 : -0.000

- The model achieved an **RMSE of 19.9** percentage points, indicating **substantial prediction error** when environmental information is ignored.
- The **MAE of 13.2 percentage points** shows the **average deviation from observed bleaching values**.
- An **R^2 value close to zero** confirms that the model explains **virtually none of the variance** in bleaching severity.
- These results establish a minimum performance threshold that all subsequent regression models must exceed to demonstrate meaningful predictive skill.

OLS Regression

Model performance: $R^2 = 0.12$ (RMSE = 18.7, MAE = 11.7)

→ OLS explains a modest proportion of variation in bleaching severity.

Statistical significance: F-statistic = 326.1, $p < 0.001$

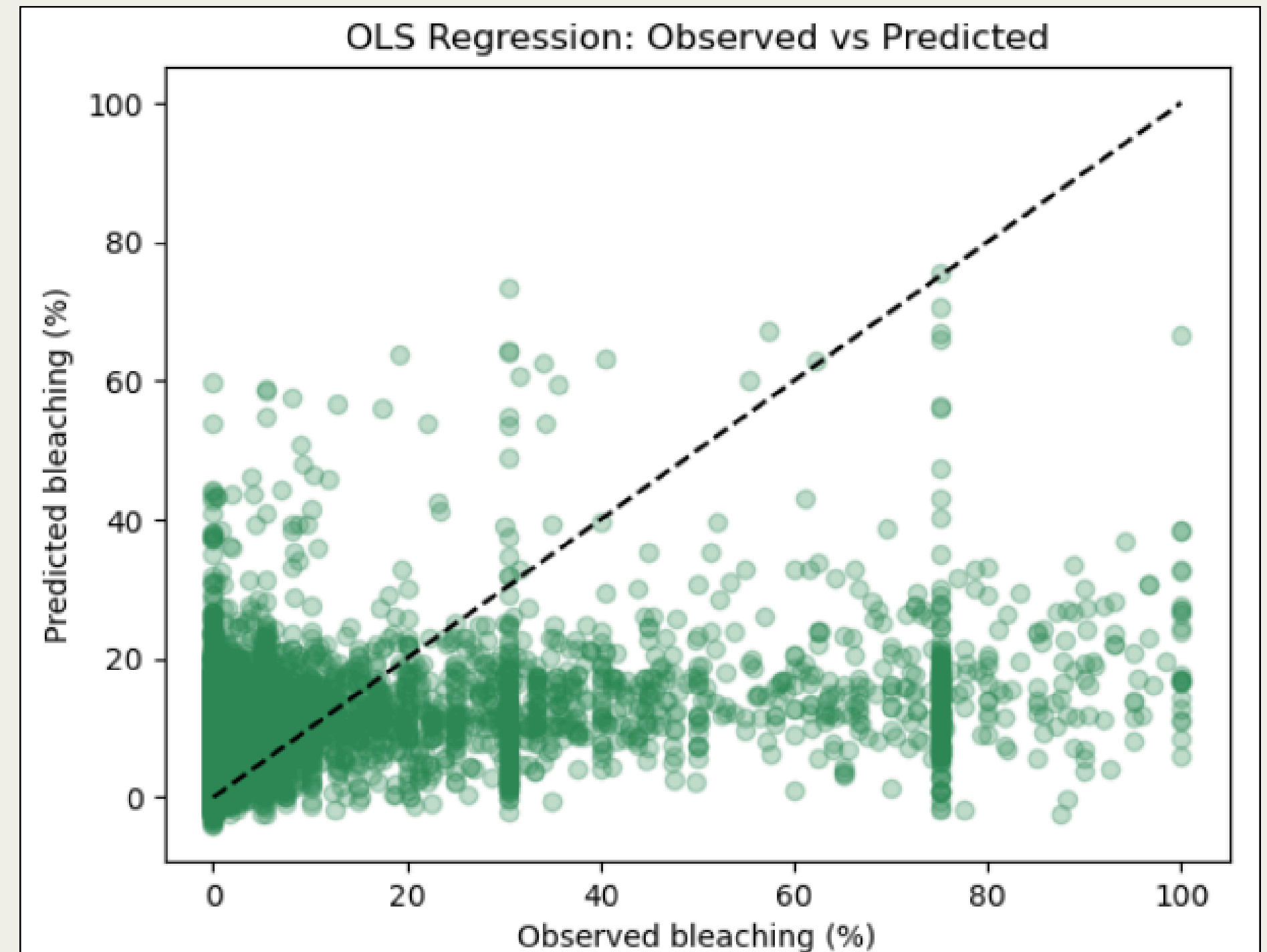
→ Predictors are collectively significant despite low explanatory power.

Key drivers:

→ Degree Heating Weeks (DHW) is the strongest positive predictor.

→ Turbidity and maximum SSTA are not significant in this linear framework.

Model limitations: Observed vs predicted plot shows weak agreement and systematic under-prediction of severe bleaching, indicating non-linear dynamics not captured by OLS.



Random Forest Regression

Strong predictive performance: RMSE = 11.6, MAE = 5.3, $R^2 = 0.66$

Substantial improvement over baseline:

- Error reduced by ~40% compared to the dummy model
- Explains ~66% of variance in bleaching severity

Key insight:

- Indicates bleaching severity is driven by non-linear relationships and interactions between environmental factors that linear models cannot capture

Although Random Forest models perform well out-of-the-box, **hyperparameter tuning was conducted to assess whether further performance gains could be achieved** and to confirm that the model was operating near its optimal capacity. I tuned the model using **RandomizedSearchCV with 5-fold cross-validation**, which efficiently tests multiple parameter combinations while limiting computational cost. Parameters such as the number of trees, tree depth, minimum leaf size, and number of features per split were explored.

Tuned Random Forest

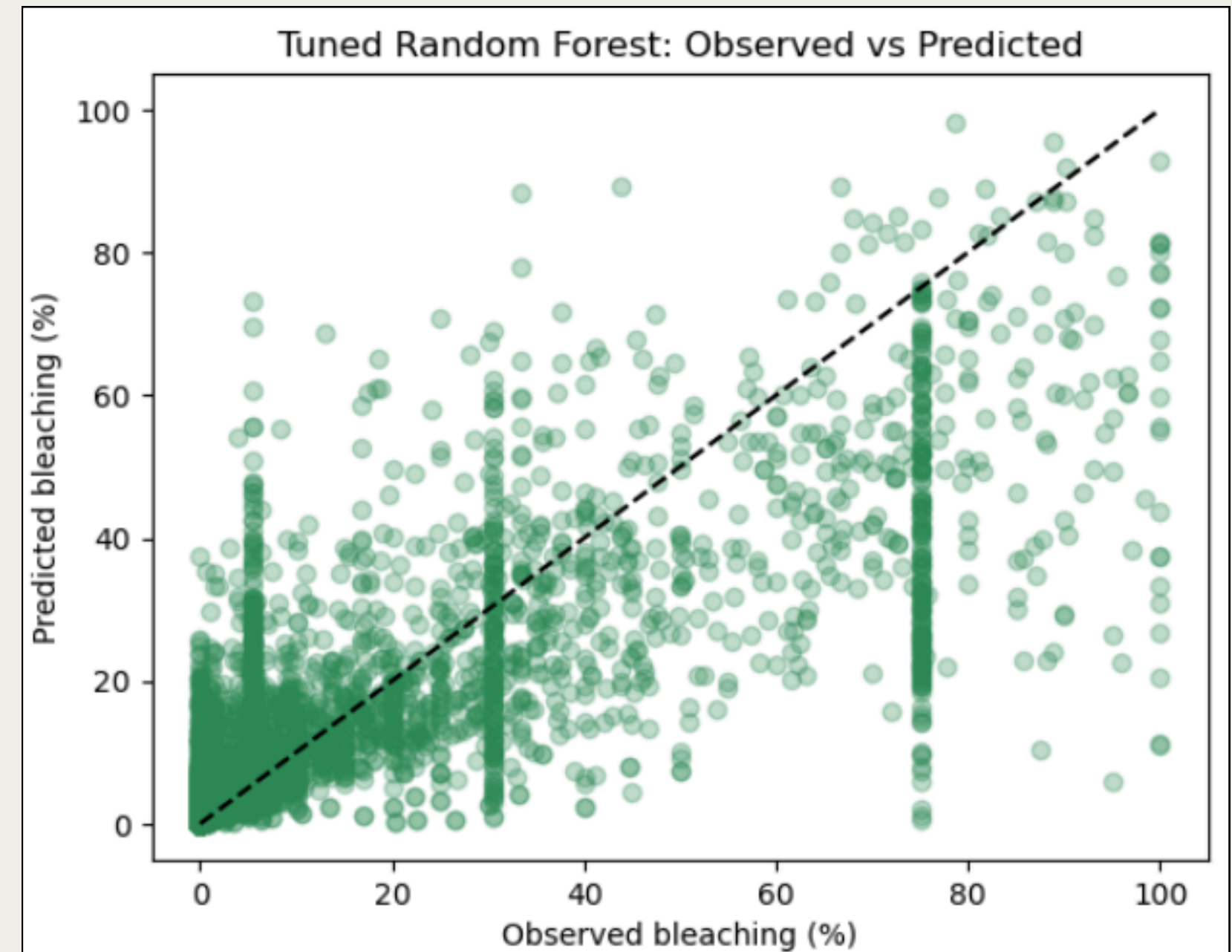
Tuning impact and generalisation

- Minimal performance change with tuning (R^2 : 0.663 $\rightarrow$ 0.666), indicating the model captures most predictive signal without heavy optimisation.
- Validation RMSE (CV): 12.22 and test RMSE: 11.51 are closely aligned, suggesting stable generalisation and low overfitting risk.

Observed vs predicted patterns

- Good agreement for low-to-moderate bleaching levels.
- Increasing spread and uncertainty at high bleaching values.
- Extreme bleaching events remain systematically under-predicted.

Overall, Random Forest substantially outperforms simpler models and is well-suited for non-linear bleaching dynamics, though uncertainty remains for rare, extreme events.



XGBoost Regression

Strong non-linear performance: RMSE = 13.2, MAE = 7.4, $R^2 = 0.56$

Improvement over linear regression:

- Explains ~56% of variance in bleaching severity
- Substantially lower error than OLS, though weaker than Random Forest

Key insight:

- Confirms coral bleaching is driven by complex, non-linear environmental processes better captured by gradient-boosted trees than linear models

Since XGBoost performance is highly sensitive to hyperparameters controlling tree complexity, learning rate, and regularisation, hyperparameter tuning was applied to optimise predictive performance and improve generalisation.

Tuned XGBoost

Performance and generalisation

- Validation RMSE (13.11) closely aligns with test RMSE (12.61), indicating good generalisation and low overfitting risk.
- Test performance (MAE = 6.95, $R^2 = 0.599$) shows strong improvement over linear models, though below Random Forest.

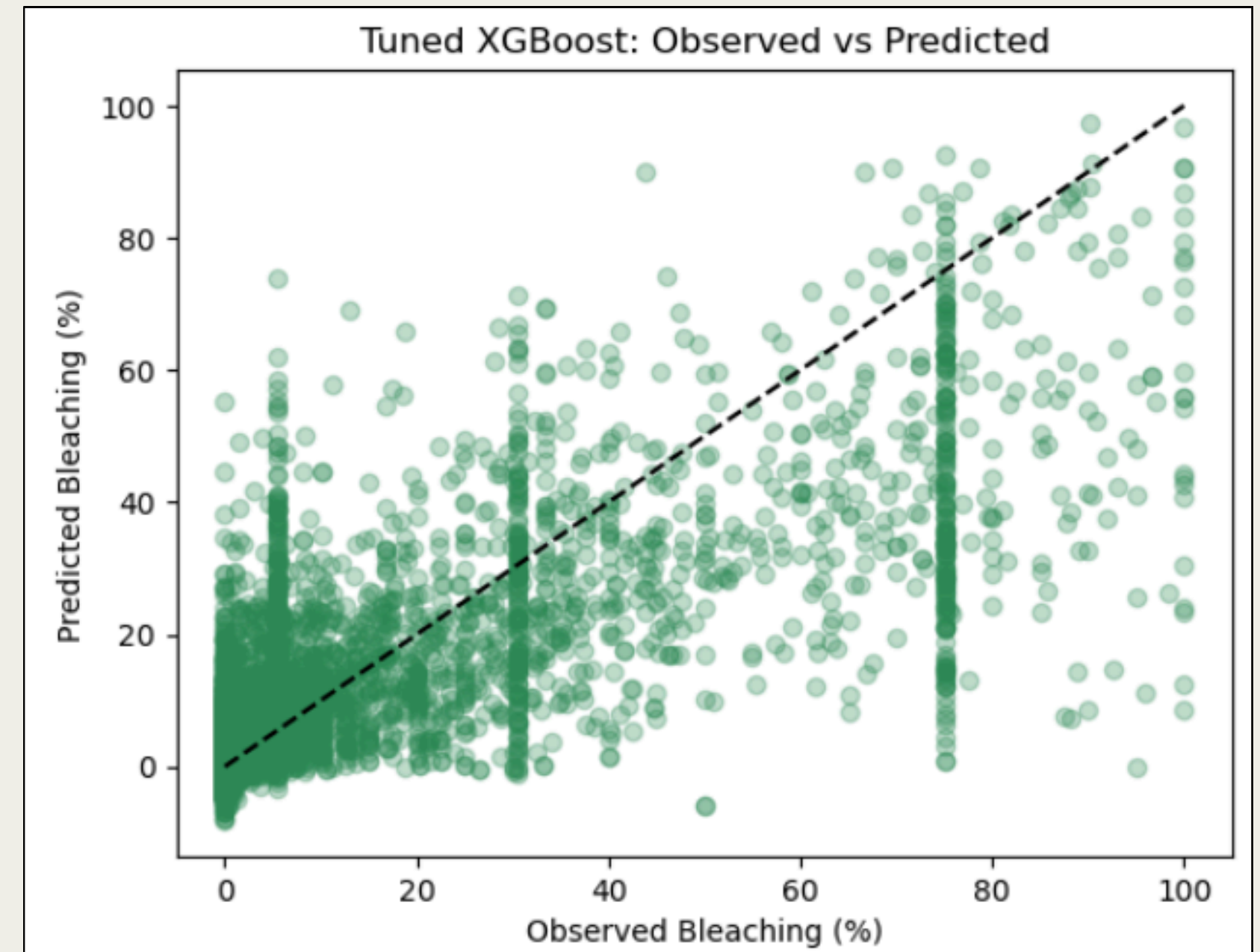
Tuning impact

- Tuning reduced error and increased R^2 .
- Greater gains than Random Forest, indicating higher sensitivity to hyperparameter settings.

Observed vs predicted patterns

- Improved alignment at low–moderate bleaching levels.
- Under-prediction and uncertainty persist for extreme events.

Overall, the tuned XGBoost captures non-linear bleaching patterns effectively but remains limited for rare severe events.



Model evaluation

- Model comparison shows that **bleaching severity is dominated by non-linear relationships**, with linear approaches capturing only a small fraction of the underlying signal.
- **Tree-based ensemble models (Random Forest and XGBoost) are far more effective at explaining global bleaching patterns**, indicating strong interactions between environmental drivers.
- The **tuned Random Forest model performs best overall**, suggesting that increased complexity beyond this point delivers diminishing returns for this dataset.
- Persistent under-prediction of extreme bleaching across all models **highlights remaining uncertainty and the influence of local or unmeasured factors**.

Model	RMSE	MAE	R ²
Dummy Regressor	19.91	13.15	-0.000
OLS Regression	18.66	11.71	0.121
Tuned Random Forest	11.51	5.45	0.666
Tuned XGBoost	12.61	6.95	0.599

→ Results support using non-linear models for risk screening and monitoring, complemented by region-specific data and informed judgement for decision-making around high-impact events.

SHAP analysis

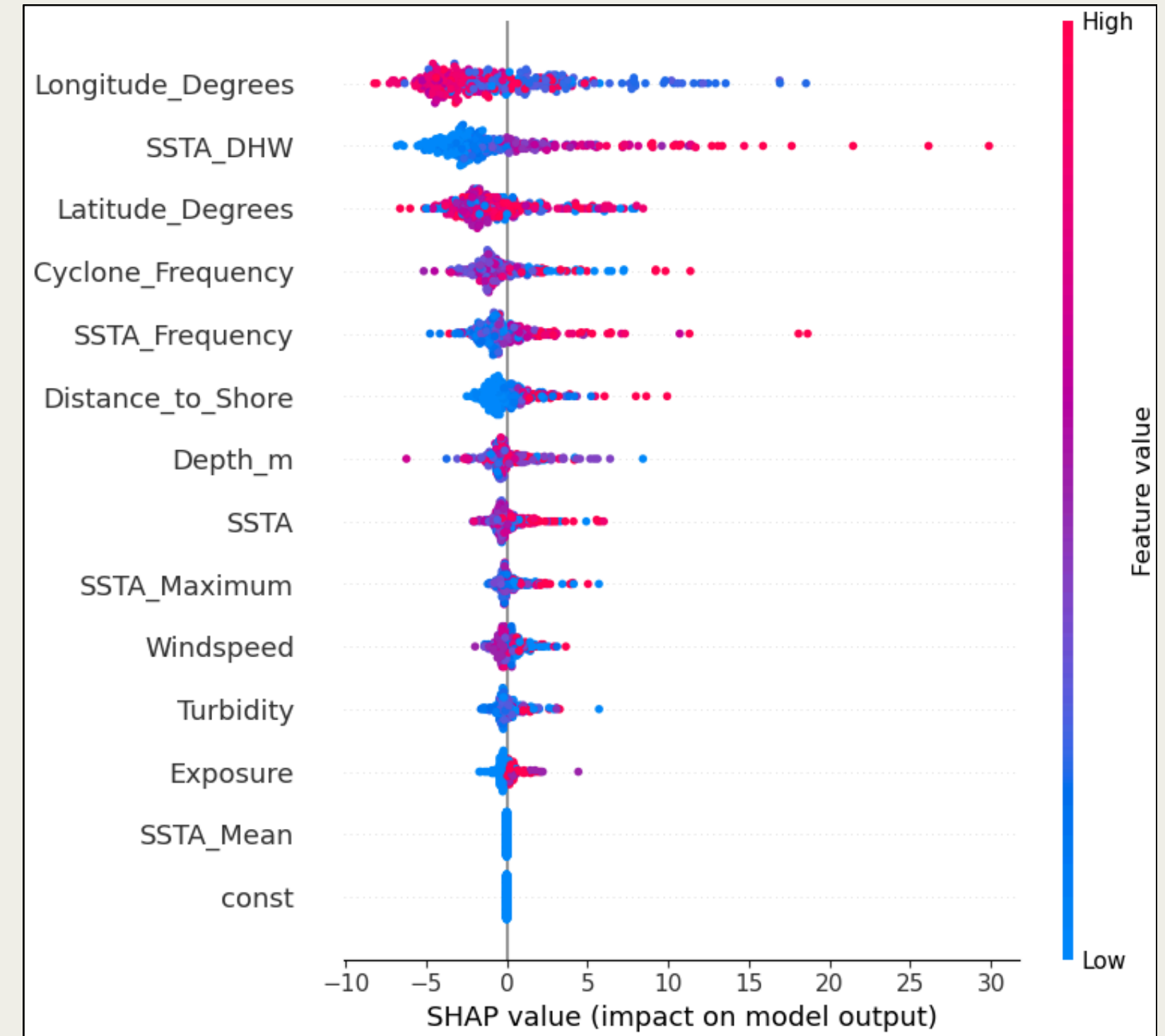
Thermal stress is the dominant driver of bleaching risk. Degree Heating Weeks (DHW) and related temperature metrics consistently exert the largest influence on predicted bleaching severity, confirming heat stress as the primary global driver.

Bleaching responses vary strongly by location. Latitude and longitude have a substantial impact on predictions, indicating important regional differences in vulnerability and exposure rather than a uniform global response.

Environmental stressors act in complex, non-linear ways. Factors such as cyclone frequency, distance to shore, and temperature variability influence bleaching differently depending on context, reinforcing that risk cannot be captured by simple linear relationships.

Some factors add limited additional explanatory power at the global scale. Variables such as turbidity and exposure contribute less to predictions once thermal and spatial drivers are accounted for, though they may remain important locally.

Overall, model behaviour is scientifically credible. The model's learned patterns align with established coral bleaching mechanisms, supporting its use for large-scale risk assessment while highlighting ongoing uncertainty for extreme bleaching events.



Limitations

Uneven spatial coverage and detection bias

Coral bleaching observations are unevenly distributed across ocean regions, with well-studied reefs far more represented than remote or inaccessible areas. Severe and visually obvious bleaching events are also more likely to be recorded than mild or short-lived events. Together, these biases mean that some regions and event types are likely under-represented.

Data quality and methodology

Bleaching records come from multiple sources that differ in survey methods, observer expertise, and reporting standards. Definitions of bleaching severity and monitoring protocols also vary across regions and have evolved over time. This limits direct comparability between regions and reduces confidence in fine-scale severity estimates.

Temporal bias and reporting changes over time

Earlier years are likely underrepresented due to less systematic monitoring and fewer long-term datasets. Observations tend to cluster around major heatwave events, which may exaggerate apparent temporal trends. As a result, increases in reported bleaching may partly reflect improved detection rather than solely increased impact.

Recommendations

1) Prioritise region-specific management strategies

Ocean-level analysis shows substantively different bleaching regimes across regions. Management and conservation actions should therefore be tailored to ocean basins or regional reef systems, rather than relying on uniform global interventions.

2) Focus mitigation efforts on reducing accumulated thermal stress

Across all models, cumulative heat-stress metrics (e.g. Degree Heating Weeks) were consistently stronger drivers of bleaching severity than single peak temperature anomalies. Mitigation, early-warning systems, and reef monitoring should prioritise the duration and frequency of marine heat stress, not just maximum temperature thresholds.

3) Incorporate extreme-event metrics into risk assessments

Average bleaching levels alone mask vulnerability to severe episodic events. Regions with similar mean bleaching can experience very different upper-quantile impacts. Risk assessments should therefore include extreme-event indicators (e.g. upper-percentile bleaching or exceedance probabilities) alongside mean values to better inform preparedness and response.

4) Use non-linear models to support evidence-based decision-making

Tree-based ensemble models substantially outperformed linear regression, indicating that coral bleaching dynamics involve non-linear interactions. For applied assessments, non-linear models should be used to support risk screening and prioritisation, with simpler linear models retained primarily for interpretability and communication.

Next steps / improvements

Target extreme events more explicitly: Severe bleaching is rare and under-predicted. Event-focused approaches (e.g. weighting extreme cases or threshold-based modelling) could improve sensitivity where impacts are greatest.

Incorporate temporal context: Adding lagged or cumulative thermal stress metrics may better capture compounding effects that precede severe bleaching.

Enhance local and regional detail: Integrating higher-resolution or reef-specific variables (e.g. local oceanography or ecosystem characteristics) could improve predictions beyond global drivers.

Explore alternative modelling approaches: Quantile-based or hybrid models may better represent uncertainty and tail behaviour associated with extreme bleaching.

Overall, future work should prioritise improving detection and interpretation of high-impact bleaching events rather than marginal gains in average accuracy.

Appendix

OLS regression coefficients

Variable	Coefficient
Depth_m	+0.47
Windspeed	+0.33
Exposure	+1.41
SSTA	+0.27
SSTA_Frequency	+0.31
SSTA_DHW	+1.00

Random Forest & XGBoost tuned vs untuned

Model	RMSE	MAE	R ²
Random Forest	11.56	5.31	0.663
Tuned Random Forest	11.51	5.45	0.666
XGBoost	13.24	7.35	0.558
Tuned XGBoost	12.61	6.95	0.599